

audit_portal_dashboard

User Instructions

Overview

`audit_portal_dashboard` is a Flask-based web application used to organize and review manufacturing accountability records.

The portal extends the local manufacturing workflow into a: - web-based - multi-user - remote auditing

environment.

The portal allows: - manufacturers to upload manufacturing records - auditors to review and verify batches - remote audit workflows - centralized report management

The portal is intended as a demonstration of: - voluntary accountability - decentralized manufacturing trust systems - cryptographic audit workflows - remote verification concepts

Launching the Flask Server

Before using the portal, the Flask server must be started locally.

Open a Terminal

Navigate to the:

`audit_portal_dashboard`

folder.

Example:

`Secure Slicer/audit_portal_dashboard/`

Activate the Python Environment

Example:

`.venv\Scripts\activate`

or:

```
python -m venv .venv
```

if the environment has not yet been created.

Install Requirements

If needed:

```
pip install -r requirements.txt
```

Start the Server

Run:

```
python app.py
```

The Flask server should start and display a local address similar to:

```
http://127.0.0.1:5000
```

Opening the Portal

Open a web browser and navigate to:

```
http://127.0.0.1:5000
```

The login page should appear.

Logging In

The portal contains separate user roles.

Example demonstration accounts:

```
manufacturer1 / demo123  
manufacturer2 / demo123  
auditor1 / demo123
```

Manufacturer Accounts

Manufacturer accounts are intended for: - uploading manufacturing records - uploading audit bundles - reviewing audit reports

Manufacturers normally interact only with their own facility records.

Auditor Accounts

Auditor accounts are intended for: - reviewing uploaded batches - selecting audit bundles - running verification - reviewing audit reports

Auditors can access records across multiple facilities.

Dashboard Navigation

After logging in, the portal opens to the main dashboard.

The dashboard displays: - uploaded manufacturing records - facility and instrument structure - recent reports - audit summaries

Navigation links appear at the top of the page.

Examples:

Dashboard
Manufacturer
Auditor
Logout

Available options depend on the user role.

Manufacturer Dashboard

Manufacturer users can: - upload public batch files - upload audit bundles - review uploaded records - review generated audit reports

Uploading Public Batch Files

Public batches are generated by:

`2-merkle_batch_generator.py`

Example batch filename:

`batch_ForgeWorks_Printer01_9f2a1c4d.json`

Upload Steps

1. Open:
`Manufacturer Dashboard`
2. Locate:
`Upload Public Batch`
3. Enter the:
`Instrument ID`
4. Press:
`Choose File`
5. Select the batch JSON file
6. Press:
`Upload Batch`

The batch becomes available for remote verification.

Uploading Audit Bundles

Audit bundles are generated by:

`4-build_audit_bundle_gui.py`

Example:

`audit_bundle_ForgeWorks_Printer01.zip`

Upload Steps

1. Open:
`Manufacturer Dashboard`
2. Locate:
`Upload Audit Bundle`
3. Enter the:
`Instrument ID`
4. Press:

Choose File

5. Select the audit ZIP file

6. Press:

Upload Audit Bundle

The audit bundle becomes available for local or remote verification.

Finding Uploaded Files

Uploaded records appear in the:

Your Uploads

section.

This section displays: - upload type - instrument ID - original filename - upload date

Finding Audit Results (Manufacturers)

Manufacturers can review audit results in the:

Your Reports

section.

This section displays: - report creation date - instrument ID - PASS or FAIL status - report links

Press:

view

to open the full verification report.

Auditor Dashboard

Auditor accounts can: - review uploaded manufacturing batches - review uploaded audit bundles - run verification - review verification reports

The Auditor Dashboard displays: - grouped facility records - uploaded batches - uploaded audit bundles - saved verification reports

Running Remote Verification

Auditors perform verification using: - a public batch - an optional audit bundle

Verification Steps

1. Open:

`Auditor Dashboard`

2. Locate:

`Run Verification`

3. Select:

`Public Batch`

4. Optionally select:

`Audit Bundle`

5. Press:

`Run Verification`

The portal performs: - public batch validation - Merkle proof checks - local bundle comparison - manufacturing consistency verification

Verification Reports

After verification completes, a report is generated.

Reports may include: - PASS or FAIL status - Merkle validation - bundle comparison - matched jobs - missing jobs - leaf hash mismatches - warnings - findings

Reports are stored inside the portal database and can be reopened later.

Local Verification vs Remote Verification

The standalone script:

`5-verify_audit_batch.py`

is primarily intended for: - local verification - offline demonstrations - direct file review

The web-based:

`audit_portal_dashboard`

extends the same concepts into: - remote auditing - multi-user workflows - uploaded manufacturing records - centralized report management

Both systems use the same underlying cryptographic accountability concepts.

Recommended Workflow

Typical workflow:

1. Generate manufacturing jobs using:

`1-merkle_integrated_slicer.py`

2. Generate manufacturing batches using:

`2-merkle_batch_generator.py`

3. Optionally publicly anchor batches using:

`3-merkle_github_anchor.py`

4. Build audit bundles using:

`4-build_audit_bundle_gui.py`

5. Upload:

- public batches
- audit bundles

into:

`audit_portal_dashboard`

6. Run remote verification using auditor accounts

7. Review generated verification reports
-

Purpose of the Demonstration

The portal demonstrates: - remote manufacturing verification - decentralized accountability workflows - cryptographic audit continuity - collaborative manufacturing trust systems - multi-user audit concepts

The project is intended to encourage exploration of: - voluntary accountability - tamper-evidence systems - remote manufacturing verification - distributed trust frameworks